Width-uniform Bayesian generalization above the mean-field scaling

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Abstract

A neural-network learner can be much larger than the network needed to represent its target. We prove that, for a family of Gaussian weight priors, this excess width does not increase the number of examples that suffice for learning. The model is a fully connected network with bounded Lipschitz hidden activations and a linear readout. The prior retains only networks whose outputs have absolute value greater than $1/2$ on the entire Boolean input cube, and learning conditions on noiseless labels. Internal weight variance is $N^{-\gamma}$ at width N, with corresponding input and readout scalings. For every fixed depth and every fixed $\gamma > 1$, the sample bound is polynomial in the input dimension and the target's bounded-weight network complexity, uniformly at every width at least that complexity. The theorem holds at the specified positive variances, without taking an infinite-width or zero-variance limit. It controls both posterior average error and the fraction of inputs receiving correct-label probability at most $2/3$. The proof compares the prior mass of a small target with the total mass of functions lacking a small description, using a Gaussian neighborhood construction, singular-value bounds, and network compression. We also relate normalized negative log prior mass to bounded-weight network complexity. The polynomial degree may depend on depth and on γ; a bound polynomial in depth is not established here.

1 Introduction

Suppose a Boolean function can be computed by a small neural network, but the learner uses a much wider network. Can a Gaussian weight prior still favor that simple target strongly enough to learn it from few examples? A bound based only on the learner's number of parameters cannot answer this question: that number grows with width even when the target is unchanged. The desired guarantee instead depends on the size of a small implementation of the target, uniformly over the learner's excess width.

We prove such a guarantee for noiseless Bayesian classification. The learner places independent Gaussian priors on its weights, restricts this prior to networks with an absolute output margin on every Boolean input, and then conditions on agreement with the training labels. The prior family is indexed by an exponent γ: the input, internal, and readout variances are respectively $N^{1-\gamma}/d$, $N^{-\gamma}$, and $N^{-\gamma-1}$. The value $\gamma = 1$ is the mean-field scaling in this convention; $\gamma > 1$ makes all three variances smaller by the common factor $N^{1-\gamma}$.

Our main result is that, for every fixed depth L and fixed $\gamma > 1$, this learner has sample complexity polynomial in the input dimension d, the target complexity C, the inverse desired error,

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and the logarithm of the inverse failure probability, for every width $N \geq C$. Here $C = C_L(f)$ is the smallest width of a depth-L implementation of the target that has strict output margin greater than $1/2$ and weights bounded in absolute value by C. The training and test inputs may follow any common distribution on the Boolean cube. With sufficiently many examples, the posterior assigns the correct label probability greater than $2/3$ on all but the desired fraction of test inputs; it also has small average label error. The precise statement is Theorem 4.3, together with the confidence remark following its proof.

The width is finite and can be as small as the target complexity. The variance is the specified nonzero value at that width; it is not sent to zero after fixing the network. Depth, however, is held fixed: the degree of the sample polynomial may depend on L and γ. The theorem also concerns the exact Bayesian posterior, without an assertion that it can be computed efficiently.

Why the prior learns small targets. For a Bayesian learning bound, it is not enough to exhibit one inexpensive set of target weights. We need to compare the probability of the target with the probability of competing functions. The proof has three steps. First, replicate the neurons of a small target network to embed it at any larger width, and lower-bound the Gaussian probability of a neighborhood that still computes the target. Second, use $\gamma > 1$ to show that, apart from an event of small total probability, a margin-separated draw has a low-norm implementation after truncating small singular values. Third, compress every such implementation into a network whose description length is independent of learner width. A finite-class argument then excludes inaccurate functions using the training labels. A direct target-probability bound covers the smaller widths where the spectral estimate is unavailable.

This develops the strong-prior argument in Vaintrob and Hänni's lecture notes [1] and the finite-γ draft [2]. The presentation includes the model, probability bounds, and complete compression argument; no companion paper is needed for the learning proof. The external tools are standard Gaussian concentration and isoperimetry. Section 5 on prior mass also makes precise a relation between function probability and bounded-weight network complexity.

2 Model and prediction convention

Fix $d \geq 1$, $L \geq 2$, and $\mathcal{X} = \{-1, 1\}^d$. Let $\bar{x} = (1, x_1, \ldots, x_d)$, and let $\phi : \mathbb{R} \to [-1, 1]$ be 1-Lipschitz. The architecture has $L - 1$ hidden layers of common width N and a linear readout:

$$h^0(x) = \bar{x}, \qquad h^s(x) = \phi(W^s h^{s-1}(x)) \quad (1 \leq s < L), \qquad z_w(x) = W^L h^{L-1}(x).$$

There are no additional biases beyond the constant input coordinate. Tanh satisfies the activation assumptions. Non-polynomiality is not needed for the theorem when the target is assumed representable.

For a Boolean function g, set

$$U_{N,g} = \{w : g(x) z_w(x) > 1/2 \text{ for every } x \in \mathcal{X}\}, \qquad \mathcal{H}_N = \{g : U_{N,g} \neq \varnothing\}, \qquad U_N = \bigcup_{g \in \mathcal{H}_N} U_{N,g}.$$

Define

$$\kappa_N(w) = \frac{d}{N}\|W^1\|_F^2 + \sum_{s=2}^{L-1} \|W^s\|_F^2 + N\|W^L\|_2^2, \qquad \kappa_N(g) = \inf_{w \in U_{N,g}} \kappa_N(w).$$

Let $C_L(g)$ be the smallest integer $C \geq 1$ for which a width-C network in $U_{C,g}$ has every weight bounded in absolute value by C. We assume $C_L(f) < \infty$ for the target. This is bounded-weight

network complexity at fixed depth; it does not silently identify arbitrary Boolean circuit size with fixed-depth network size.

Under the raw law $\mu_{N,\gamma}$ all weights are independent centered Gaussians with standard deviations

$$\sigma_1 = N^{(1-\gamma)/2}/\sqrt{d}, \qquad \sigma_s = N^{-\gamma/2}\ (2 \le s < L), \qquad \sigma_L = N^{-(\gamma+1)/2}.$$

The density is proportional to $\exp[-N^\gamma \kappa_N(w)/2]$. The function prior is the raw law conditioned on U_N:

$$\pi_{N,\gamma}(g) = \frac{\mu_{N,\gamma}(U_{N,g})}{\mu_{N,\gamma}(U_N)}.$$

Given $S_n = ((x_i, f(x_i)))_{i=1}^n$ with independent $x_i \sim D$, the posterior $\Pi(\cdot \mid S_n)$ restricts $\pi_{N,\gamma}$ to functions satisfying all observed labels. For any distribution D on $\mathcal{X}$, write

$$\mathrm{err}_D(g) = \Pr_{x\sim D}\{g(x) \ne f(x)\}, \qquad R_D(\Pi) = \mathbb{E}_{g\sim\Pi}\,\mathrm{err}_D(g).$$

The global margin condition applies before training, including at unobserved inputs. It is part of the learner, and differs from imposing margins only on the training sample. The theorem controls the posterior probability of an incorrect label, averaged over a fresh input.

3 A cylinder construction valid throughout $\gamma \ge 1$

The following construction bounds the target's prior cost at every width $N \ge C$, without a width threshold diverging as $\gamma \downarrow 1$.

Proposition 3.1 (Uniform cloning mass bound)**.** *Fix $L \ge 2$ and a bounded 1-Lipschitz activation. There is a function $Q_L(d, C)$ bounded by a polynomial such that for every $\gamma \ge 1$, every Boolean target with $C_L(f) \le C$, and every $N \ge C$,*

$$-\frac{2}{N^\gamma}\log \Pr\{f(x)z_w(x) > 1/2 \text{ for all } x\} \le Q_L(d, C).$$

The prior variances and absence of later-layer biases are exactly those in the main model.

Proof. Choose a width-C teacher, increase its readout by a factor of three, and write its hidden matrices as B^s and readout as v. Thus the teacher has margin strictly greater than $3/2$, hidden entries bounded by C, and readout entries bounded by $3C$. If the original witness has smaller width, first use its actual width in the construction below; all estimates only improve. Partition $\{1, \dots, N\}$ into C groups of sizes k_a differing by at most one. Then

$$1 \le k_a, \qquad \frac{\max k_a}{\min k_a} \le 2, \qquad \frac{N}{k_a} \le 2C.$$

Let $u_a = k_a^{-1/2}\mathbf{1}_{\text{group } a}$ and let P be the orthogonal projection onto their span. The exact clone w_* has first-layer row B^1_b for a neuron in group b, hidden entry B^s_{ba}/k_a from group a to group b, and readout v_b/k_b on group b. Every hidden activation of the exact clone is group-constant. Moreover,

$$\|W^s_*\|_{\mathrm{op}} \le \sqrt{2}C^2 \quad (2 \le s < L), \qquad \sqrt{N}\|W^L_*\|_2 \le 3\sqrt{2}C^2.$$

The first bound follows by restricting to the group-constant subspace: the resulting matrix has entries $\sqrt{k_b/k_a}B^s_{ba}$, and its Frobenius norm is at most $\sqrt{2}C^2$.

Set $t = N^{(1-\gamma)/2} \le 1$. The random coordinates $\sqrt{d}W^1_{ij}$, $\sqrt{N}W^s_{ij}$ for $2 \le s < L$, and $\alpha_j = NW^L_j$ are independent centered Gaussians of standard deviation t. Impose the following cylinder event $\mathcal{C}_\rho$, for $0 < \rho \le 1$:

- every $\sqrt{d}W^1_{ij}$ is within ρt of $\sqrt{d}(W^1_*)_{ij}$;
- for every hidden row j and group a, the coordinate $\langle\sqrt{N}W^s_{j,\cdot}, u_a\rangle$ is within ρt of its cloned value $\sqrt{N/k_a}B^s_{ba}$;
- every α_j is within ρt of its cloned value Nv_b/k_b.

All these constraints concern independent scalar Gaussian coordinates. The orthogonal coordinates of each internal row remain independent prior Gaussians. Consequently write

$$W^s = W^s_* + R^s + E^s, \qquad R^s = R^s P, \quad E^s P = 0.$$

On the cylinder, $\|R^s\|_{\mathrm{op}} \le \|R^s\|_F \le \rho t\sqrt{C}$. The free matrix is distributed as $(t/\sqrt{N})G^s(I-P)$, with G^s a standard Gaussian matrix, independently of the cylinder coordinates. Let

$$b_L = 2 + \sqrt{2\log(2L)}.$$

The Gaussian operator-norm bound and a union bound imply, conditionally on $\mathcal{C}_\rho$, that all $\|E^s\|_{\mathrm{op}} \le b_L t$ with probability at least $1/2$. This is also valid when $L = 2$, where there are no such matrices.

Let $e_s(x) = N^{-1/2}\|h^s_w(x) - h^s_*(x)\|_2$. The input constraints give $e_1(x) \le \rho(d+1)/\sqrt{d} \le \rho(d+1)$. Since $E^s h^{s-1}_* = 0$, boundedness and Lipschitz continuity give, simultaneously for every input,

$$e_s(x) \le (\sqrt{2}C^2 + \sqrt{C} + b_L)e_{s-1}(x) + \rho\sqrt{C}.$$

Put $A = 2 + 3C^2 + b_L$. Induction yields $e_s(x) \le \rho(d + L + 1)A^{L-2}$ for $s < L$. At the output,

$$|z_w(x) - z_*(x)| \le 3\sqrt{2}C^2 e_{L-1}(x) + \rho.$$

Hence choosing

$$\rho = \big[2 + 9C^2(d + L + 1)A^{L-2}\big]^{-1}$$

ensures output discrepancy at most $1/2$. All networks in the cylinder with the operator-norm event therefore implement f with margin greater than one.

It remains to price the cylinder. Its number of constrained coordinates is

$$m = N\big[d + 2 + (L-2)C\big].$$

The sum of their squared centers is at most NB_0, where one may take

$$B_0 = d(d+1)C^2 + 2(L-2)C^4 + 36C^4.$$

Indeed, a first-layer center has magnitude at most $\sqrt{d}C$, an internal center at most $\sqrt{2C}C$, and a readout center at most $6C^2$. For $Z \sim N(0, t^2)$, every real a, and every $\rho > 0$,

$$\Pr\{|Z - a| \le \rho t\} \ge \frac{2\rho}{\sqrt{2\pi}} \exp\{-a^2/t^2 - \rho^2\}.$$

Multiplication and $t^{-2} = N^{\gamma-1}$ show

$$-\log \Pr(\mathcal{C}_\rho) \le N^\gamma B_0 + N[d + 2 + (L-2)C] \log \frac{e\sqrt{2\pi}}{2\rho}.$$

The desired probability is at least $\Pr(\mathcal{C}_\rho)/2$. Since $N \le N^\gamma$ and $N^\gamma \ge 1$, this proves the claim with

$$Q_L(d, C) = 2B_0 + \frac{4[d + 2 + (L-2)C]}{\rho} + 2.$$

Here we used $\log(e\sqrt{2\pi}/(2\rho)) \le 2/\rho$ for $0 < \rho \le 1$. The displayed expression is a polynomial in d, C at every fixed depth. □

4 Width-independent learning at every fixed $\gamma > 1$

Let $\mathcal{A}_L(d, C) = \max\{1, Q_L(d, C)\}$, where Q_L is the bound from the preceding cylinder construction. Then $\mathcal{A}_L$ is polynomially bounded and

$$\mu_{N,\gamma}(U_{N,f}) \geq \exp\{-\mathcal{A}_L(d, C)N^\gamma/2\} \qquad (\gamma \geq 1,\ N \geq C). \tag{1}$$

We now control the *total* mass of functions whose weighted norm $\kappa_N(g)$ is large. A Gaussian geometric argument gives this control for every fixed $\gamma > 1$.

Lemma 4.1 (A Gaussian singular-value tail)**.** *Let M be an $N \times N$ matrix with independent $N(0, \sigma^2)$ entries. For $a \geq 8\sigma\sqrt{N}$, let $M_{>a}$ retain exactly the singular values of M greater than a. Then, for every $K > 0$,*

$$\Pr\{\|M_{>a}\|_F^2 > K\} \leq \exp\{-K/(8\sigma^2)\}.$$

Proof. The standard Gaussian operator-norm inequality [8] implies that the ball $B_a = \{T : \|T\|_{\mathrm{op}} \leq a/2\}$ has Gaussian probability at least $1 - e^{-2N} > 1/2$. Indeed,

$$\Pr\{\|M\|_{\mathrm{op}} > 2\sigma\sqrt{N} + \sigma t\} \leq e^{-t^2/2},$$

and $a/2 \geq 4\sigma\sqrt{N}$. The Frobenius distance to B_a is

$$D(M) = \mathrm{dist}_F(M, B_a) = \left[\sum_j (s_j(M) - a/2)_+^2\right]^{1/2}.$$

The Gaussian isoperimetric inequality [6, 7], applied to the Euclidean enlargement of B_a in the N^2 matrix coordinates, gives

$$\Pr\{D(M) > t\} \leq 1 - \Phi(t/\sigma) \leq e^{-t^2/(2\sigma^2)}, \qquad t > 0,$$

where Φ is the standard normal distribution function. If $s > a$, then $s \leq 2(s - a/2)$, so $\|M_{>a}\|_F \leq 2D(M)$. Apply the last bound with $t = \sqrt{K}/2$. □

Lemma 4.2 (Most margin-separated draws have low-norm descriptions)**.** *Fix $K \geq 1$, and set*

$$a_K = \frac{1}{8L(\sqrt{K} + 1)^{L-2}}, \qquad J_K = (L + 3)K.$$

Suppose $\gamma > 1$ and

$$N^{\gamma-1} \geq 8(d + 1), \qquad N^{\gamma-1} \geq 64a_K^{-2}. \tag{2}$$

For $\mathcal{S}_N(K) = \{g \in \mathcal{H}_N : \kappa_N(g) \leq J_K\}$,

$$\mu_{N,\gamma}\left(\bigcup_{g \notin \mathcal{S}_N(K)} U_{N,g}\right) \leq Le^{-KN^\gamma/8}. \tag{3}$$

Proof. Let G_K be the event

$$\frac{d}{N}\|W^1\|_F^2 \leq K, \qquad N\|W^L\|_2^2 \leq K, \qquad \|(W^s)_{>a_K}\|_F^2 \leq K \quad (2 \leq s < L).$$

The first and last weighted norms have distributions $N^{-\gamma}\chi^2_{N(d+1)}$ and $N^{-\gamma}\chi^2_N$. For $X \sim \chi^2_D$ and $u \geq 8D$, the exponential-moment bound at parameter $3/8$ gives

$$\Pr\{X > u\} \leq 4^{D/2} e^{-3u/8} \leq e^{-u/4}.$$

The first condition in (2) therefore bounds each of these two failure probabilities by $e^{-KN^\gamma/4}$. For each internal layer, the second condition gives $a_K \geq 8N^{(1-\gamma)/2}$, so Lemma 4.1 bounds its failure probability by $e^{-KN^\gamma/8}$. A union bound yields

$$\mu_{N,\gamma}(G_K^c) \leq Le^{-KN^\gamma/8}. \tag{4}$$

Fix a draw $w \in G_K$. Replace each internal matrix by its $> a_K$ singular-value truncation, obtaining $\widetilde{w}$. The retained matrices have operator norms at most $\sqrt{K}$; the discarded matrices have operator norms at most a_K. For any input, let e_s be the Euclidean difference of the original and truncated activations at hidden layer s. Boundedness and Lipschitz continuity of the activation give

$$e_1 = 0, \qquad e_s \leq \sqrt{K}\, e_{s-1} + a_K\sqrt{N} \quad (2 \leq s < L).$$

The readout has norm at most $\sqrt{K/N}$. Thus, when $L \geq 3$,

$$|z_{\widetilde{w}}(x) - z_w(x)| \leq a_K \sum_{j=1}^{L-2} K^{j/2} \leq a_K(L-2)K^{(L-2)/2} \leq \frac{1}{8}.$$

When $L = 2$, the networks are identical, so the same final bound holds. All these bounds hold simultaneously for every input. If $w \in U_{N,g}$, the truncated network has strict margin $3/8$ for g. Doubling its readout restores strict margin $1/2$ and gives weighted norm at most $K + (L-2)K + 4K = J_K$. Therefore every $w \in G_K \cap U_N$ implements a function in $\mathcal{S}_N(K)$. The event on the left of (3) is contained in G_K^c, and (4) proves the claim. □

The key point is that (3) bounds an entire union of function events directly. No union bound over the 2^{2^d} Boolean functions is necessary. Conditioning the prior on U_N does not affect its ratio to the target event's mass.

The finite capacity bound used below. The compression theorem (Theorem A.1, proved in Appendix A) gives a polynomially bounded $m_L(d, J+1)$ such that every function with $\kappa_N(g) \leq J$ has a width-$m = m_L(d, J+1)$ realization, with strict margin $1/2$ and all weights bounded by m. The $+1$ permits an arbitrarily near minimizer when the infimum defining $\kappa_N(g)$ is not attained. For definiteness one can take

$$\rho = \frac{2^{-L}(J+1)^{-(L-1)/2}}{12(L-1)}, \qquad m_L(d, J+1) = \max\left\{\lceil 9(J+1)\rceil, \left\lceil 256\rho^{-4}\big(d + \log(L/\rho)\big)\right\rceil\right\}.$$

Writing $P_m = m(d+2) + (L-2)m^2$, define

$$H_L(d, J) = P_m \log\left(3 + 16Lm\max\{m, d+1\}[m(m+1)]^{L-1}\right). \tag{5}$$

Then

$$\log \#\{g \in \mathcal{H}_N : \kappa_N(g) \leq J\} \leq H_L(d, J). \tag{6}$$

To see this, round each weight of the compressed realization to a grid of spacing $r = [8L\max\{m, d+1\}\{m(m+1)\}^{L-1}]^{-1}$. The rounding changes each weight by at most r and keeps weights bounded

by $m+1$. A layerwise telescoping estimate in the maximum norm bounds the output change by $L\max\{m,d+1\}r[m(m+1)]^{L-1} = 1/8$. Thus every rounded array has the same Boolean signs as its original realization. Distinct functions require distinct arrays, and the number of arrays is at most $(3+2m/r)^{P_m}$. This proves (6). At fixed L, H_L is bounded by a polynomial in its arguments.

Theorem 4.3 (All widths at fixed $\gamma > 1$)**.** *Fix $L \geq 2$ and $\gamma > 1$. There exists a polynomial $p_{L,\gamma}$ such that the following holds. Let $C = C_L(f) < \infty$, let $N \geq C$, and let the function prior and consistency posterior use the global margin conditioning specified above. For every distribution D on $\{-1,1\}^d$, every $\varepsilon, \eta \in (0,1)$, and*

$$n \geq \frac{2}{\varepsilon}\left[p_{L,\gamma}(d, C, \log(1/\varepsilon)) + \log\frac{2}{\varepsilon\eta}\right],$$

with probability at least $1-\eta$ over n independent labeled examples,

$$\mathbb{E}_{g\sim\Pi(\cdot|S_n)}\,\mathrm{err}_D(g) \leq \varepsilon.$$

The polynomial is independent of N. Its exponent may depend on γ; this theorem makes no uniform assertion as $\gamma \downarrow 1$.

Proof. Set

$$A = \mathcal{A}_L(d, C), \qquad K = 4A + 8\log(2L/\varepsilon), \qquad a = a_K,$$

and define the integer threshold

$$T = \left\lceil \max\left\{C, [8(d+1)]^{1/(\gamma-1)}, (8/a)^{2/(\gamma-1)}\right\}\right\rceil.$$

First suppose $N \geq T$. By the aggregate structural tail and the target mass bound,

$$\begin{aligned}\frac{\pi_{N,\gamma}(\mathcal{H}_N \setminus \mathcal{S}_N(K))}{\pi_{N,\gamma}(f)} &= \frac{\mu_{N,\gamma}(\bigcup_{g\notin\mathcal{S}_N(K)} U_{N,g})}{\mu_{N,\gamma}(U_{N,f})} \\ &\leq L\exp\{-(K/8 - A/2)N^\gamma\} \leq \varepsilon/2.\end{aligned}$$

The set $\mathcal{S}_N(K)$ contains f: otherwise the left side would be at least one. Its log cardinality is at most $H_L(d, J_K)$ by (6). Any particular function of true error greater than $\varepsilon/2$ is consistent with n independent examples with probability at most $e^{-n\varepsilon/2}$. A union bound shows that, if

$$n \geq \frac{2}{\varepsilon}\left[H_L(d, J_K) + \log(1/\eta)\right],$$

then, with probability at least $1-\eta$, every consistent function in $\mathcal{S}_N(K)$ has error at most $\varepsilon/2$. On every training sample, the posterior denominator is at least $\pi_{N,\gamma}(f)$, so posterior mass outside this set is at most $\varepsilon/2$. The posterior risk is therefore at most ε.

Now suppose $C \leq N < T$. Since $\mu_{N,\gamma}(U_N) \leq 1$,

$$-\log\pi_{N,\gamma}(f) \leq -\log\mu_{N,\gamma}(U_{N,f}) \leq AN^\gamma/2 \leq AT^\gamma/2.$$

For any prior with positive target mass, let B_n be posterior mass assigned to functions with true error greater than $\varepsilon/2$. The posterior denominator is at least the target's prior mass, giving

$$\mathbb{E}_{S_n} B_n \leq \frac{1}{\pi_{N,\gamma}(f)} \sum_{g:\mathrm{err}_D(g)>\varepsilon/2} \pi_{N,\gamma}(g)(1-\mathrm{err}_D(g))^n \leq \frac{e^{-n\varepsilon/2}}{\pi_{N,\gamma}(f)}.$$

Markov's inequality gives $B_n \le \varepsilon/2$ with probability at least $1-\eta$ as soon as

$$n \ge \frac{2}{\varepsilon}\left[AT^\gamma/2 + \log\frac{2}{\varepsilon\eta}\right].$$

The remaining posterior mass contributes at most $\varepsilon/2$ to risk.

It is consequently enough to majorize

$$\max\{H_L(d,(L+3)K),\ AT^\gamma/2\}$$

by $p_{L,\gamma}(d, C, \log(1/\varepsilon))$. The cylinder bound makes A polynomially bounded in d, C at fixed L; then K is polynomially bounded in $d, C, \log(1/\varepsilon)$. For fixed $L, \gamma > 1$, so are a^{-1}, T, T^γ, and the last display. This proves the stated uniform-in-width polynomial guarantee. □

For a confidence threshold $q \in (0,1)$, define

$$F_q(\Pi; f, D) = \Pr_{x\sim D}\{\Pi(g(x) = f(x) \mid S_n) \le q\}.$$

This counts an input as successful only when the posterior probability of its correct label is strictly greater than q. Since the posterior probability of an error is at least $1-q$ on the failure set,

$$F_q(\Pi; f, D) \le \frac{R_D(\Pi)}{1-q}.$$

In particular $F_{2/3} \le 3R_D(\Pi)$. Applying the theorem at accuracy $\varepsilon/3$ proves the same width-independent polynomial sample bound for having correct-label confidence greater than $2/3$ on at least a $1-\varepsilon$ fraction of test inputs.

A posterior-majority classifier has risk at most twice the posterior average risk, since every majority mistake places posterior mass at least $1/2$ on the wrong label. Applying the theorem with accuracy $\varepsilon/2$ therefore gives the same type of guarantee for majority prediction.

Comparison with the supplied draft. The supplied draft's [2] structural estimates are compatible with the aggregate argument above, but its learning section instead sums a per-function bound over all Boolean functions. That step introduces an unnecessary $N^\gamma \gtrsim 2^d$ condition. Controlling the bad parameter event directly removes this exponential width requirement. The present theorem also covers the remaining polynomial range of widths through the elementary posterior bound. Neither this repair nor the spectral lemma establishes the endpoint $\gamma = 1$: at that endpoint the Gaussian bulk operator norm remains of constant order, whereas the truncation error requires a small cutoff.

5 Prior mass and bounded-weight network complexity

The same estimates also give the mass-complexity equivalence proposed in the supplied draft [2], now without an auxiliary width proviso.

Corollary 5.1 (Polynomial equivalence of mass and network complexity)**.** *For every fixed $L \ge 2$ and $\gamma > 1$, there is a polynomial $P_{L,\gamma}$ with the following property. Define the unconditioned mass complexity*

$$\overline{\kappa}_{N,\gamma}(f) = -\frac{2}{N^\gamma}\log \mu_{N,\gamma}(U_{N,f}).$$

At every $N \ge 1$, whenever this quantity is finite,

$$C_L(f) \le P_{L,\gamma}(d, 1 + \overline{\kappa}_{N,\gamma}(f)).$$

Conversely, whenever $N \geq C = C_L(f)$,

$$\overline{\kappa}_{N,\gamma}(f) \leq Q_L(d, C).$$

The converse holds for every $\gamma \geq 1$, with the same polynomial Q_L from the cylinder construction. The first polynomial is independent of N, but may have degree depending on γ.

Proof. The converse is the cylinder mass bound. For the first direction, write $b = \overline{\kappa}_{N,\gamma}(f)$, $B = \max\{1, b\}$, and $K = 4B + 8\log(2L)$. Set $a = [8L(\sqrt{K}+1)^{L-2}]^{-1}$ and

$$T = \left\lceil \max\{[8(d+1)]^{1/(\gamma-1)}, (8/a)^{2/(\gamma-1)}\} \right\rceil.$$

This is polynomially bounded in d, B at fixed L, γ. If $N \geq T$, the proof of Lemma 4.2 gives

$$\mu_{N,\gamma}(G_K^c) \leq Le^{-KN^\gamma/8} \leq \tfrac{1}{2}e^{-bN^\gamma/2} < \mu_{N,\gamma}(U_{N,f}).$$

Thus some implementing draw is in G_K, and its truncated, rescaled network implements f with weighted norm at most $(L+3)K$. Theorem A.1 gives $C_L(f) \leq m_L(d, (L+3)K + 1)$.

If $N < T$, let $D_N = N(d+2) + (L-2)N^2$. Under the raw prior, $q = N^\gamma \kappa_N(w) \sim \chi^2_{D_N}$. For $u = 4bN^\gamma + 2D_N \log 2 + 4$, the exponential-moment bound gives

$$\Pr\{q \geq u\} \leq 2^{D_N/2} e^{-u/4} = e^{-bN^\gamma - 1} < \mu_{N,\gamma}(U_{N,f}).$$

Consequently an implementing draw has

$$\kappa_N(w) < u/N^\gamma \leq 4B + 2\big[T(d+2) + (L-2)T^2\big]\log 2 + 4 =: H.$$

Another application of Theorem A.1 gives $C_L(f) \leq m_L(d, H+1)$. Both resulting bounds are polynomially bounded in d, B at fixed L, γ, proving the claim. □

6 Scope of the result

The theorem supplies an explicit finite-variance extension of the strong-prior argument. For large enough width the relevant-class sample bound depends on $L, d, C, \log(1/\varepsilon)$ but not on γ; γ determines when the spectral cutoff lies above the Gaussian bulk. Covering all widths uses a polynomial whose degree grows as γ approaches one. This is a limitation of the present argument, rather than a lower bound asserting that such deterioration is necessary.

The cloning bound is uniform over $\gamma \geq 1$, including the endpoint. At $\gamma = 1$ the internal Gaussian matrices have a bulk of constant operator norm, so the small-cutoff argument used here does not establish a full-depth compression theorem. A separate mean-field paper [3] proves polynomial sample complexity for tanh at every width $N \geq C_L(f)$, at arbitrary fixed depth, using finite-width covariance estimates and a different compression argument. That result is not needed for the present proof. For one hidden layer, a direct chi-square argument already gives the every-width conclusion for all $\gamma \geq 1$ and any activation satisfying the assumptions of Section 2. Indeed, when $L = 2$ there are no internal matrices, and the event G_K in the proof of Lemma 4.2 consists of the two chi-square bounds. These need only $KN^\gamma \geq 8N(d+1)$, which holds at every width and every $\gamma \geq 1$ once $K \geq 8(d+1)$. Every draw in $G_K \cap U_N$ then has $\kappa_N \leq 2K$, and $\mu_{N,\gamma}(G_K^c) \leq 2e^{-KN^\gamma/4}$. With $A = \mathcal{A}_2(d, C)$ and $K = \max\{8(d+1), 2A + 4\log(4/\varepsilon)\}$, the large-width argument in the proof of Theorem 4.3 therefore applies at every $N \geq C$ and every $\gamma \geq 1$, with a sample polynomial independent of γ.

For $\gamma < 1$, tanh's diverging preactivation variance leads to a saturated, iterated-arcsine kernel. Under global margin conditioning its learning limit depends further on the sign of γ. The companion kernel note [4] treats these limits and explains why a lower bound for regression using the posterior mean cannot be substituted for a classification lower bound under the present conventions. For tanh at every fixed $\gamma < 1$, a separate paper [5] proves an $\exp(\Omega(d))$ sample lower bound for full parity under uniform training inputs, for the correct-label confidence criterion, with width taken to infinity before dimension.

A A complete norm-compression proof

The next theorem is deterministic. It applies at any width and does not refer to the prior exponent. It is the compression result used in the finite capacity bound above.

Theorem A.1 (Compression)**.** *Let $d \geq 1$, $L \geq 2$, and let $\phi : \mathbb{R} \to [-1, 1]$ be 1-Lipschitz. Suppose a width-N network W satisfies $\kappa_N(W) \leq K$, where $K \geq 1$. Set*

$$\begin{aligned} \eta &= \frac{1}{12(L-1)}, \qquad \rho = \eta 2^{-L} K^{-(L-1)/2}, \\ m_L(d, K) := M &= \max\left\{\lceil 9K \rceil, \left\lceil 256\rho^{-4}\big(d + \log(L/\rho)\big)\right\rceil\right\}. \end{aligned} \tag{7}$$

There is a width-M network $\widetilde{W}$ such that

$$\max_{x \in \{-1,1\}^d} |z_{\widetilde{W}}(x) - z_W(x)| \leq \frac{1}{12}, \qquad \kappa_M(\widetilde{W}) \leq 4K. \tag{8}$$

If W implements a Boolean function g with strict margin $1/2$, then multiplying the readout of $\widetilde{W}$ by $3/2$ gives a width-M implementation of g with strict margin $1/2$, weighted norm at most $9K$, and every weight of absolute value at most M. In particular, $C_L(g) \leq M$. For fixed L, M is bounded by a polynomial in d and K.

Proof. We first truncate large entries and then sample the hidden neurons. All norms of rows below are Euclidean, and all matrix norms with subscript F are Frobenius norms.

Truncation. For $2 \leq s \leq L$, let $r_i^s = \|W_{i,*}^s\|_2$. In each nonzero row, delete entries larger in absolute value than

$$\tau_i^s = \frac{2r_i^s}{\rho\sqrt{N}}.$$

Leave zero rows unchanged, write $\overline{W}^s$ for the resulting matrices, and set $\overline{W}^1 = W^1$. Here and below W^L is the single readout row. Because all hidden activations have absolute value at most one, deleting a row's large entries changes its preactivation by at most

$$\sum_{j:\,|W_{ij}^s| > \tau_i^s} |W_{ij}^s| \leq \frac{(r_i^s)^2}{\tau_i^s} = \frac{\rho}{2}\sqrt{N}\, r_i^s.$$

The vector of changes at an internal layer therefore has norm at most $(\rho/2)\sqrt{N}\|W^s\|_F$. Every original or truncated internal matrix has operator norm at most $\sqrt{K}$, and the readout has norm at most $\sqrt{K/N}$. The activation is a contraction in Euclidean norm. Thus truncating layer $s < L$ changes the output by at most

$$\frac{\rho}{2}\sqrt{NK}\, K^{(L-s-1)/2}\sqrt{K/N} = \frac{\rho}{2}K^{(L-s+1)/2} \leq \frac{\eta}{2}.$$

Truncating the readout changes it by at most $(\rho/2)\sqrt{K} \le \eta/2$. These bounds also hold when other layers have already been truncated. Telescoping over the $L-1$ truncations gives total output error at most $1/24$.

Independent samples and three events. For each hidden layer s, independently choose a list $I_s(1), \dots, I_s(M)$ of independent uniform indices in $\{1, \dots, N\}$. Repeated indices represent separate copies of a neuron. Define the fully sampled network by

$$\begin{aligned} \widetilde{W}^1_{a,*} &= W^1_{I_1(a),*}, \\ \widetilde{W}^s_{a,b} &= \frac{N}{M} \overline{W}^s_{I_s(a),I_{s-1}(b)} && (2 \le s \le L-1), \\ \widetilde{W}^L_b &= \frac{N}{M} \overline{W}^L_{I_{L-1}(b)}. \end{aligned}$$

For the analysis expose these independent lists in reverse order, $I_{L-1}, \dots, I_1$. At the stage sampling layer m, all earlier layers still have their original, truncated weights. Let $\overline{h}^m_j(x)$ denote their exactly computed activations. The estimated sum into a retained row i of layer $m+1$ is

$$\frac{N}{M} \sum_{b=1}^{M} \overline{W}^{m+1}_{i,I_m(b)} \overline{h}^m_{I_m(b)}(x).$$

Its expectation is the corresponding full sum over N neurons. Conditional on the later lists, its summands are independent and bounded in absolute value by $(N/M)\tau^{m+1}_i$. Hoeffding's inequality [9] shows that its deviation exceeds $(\rho/2)\sqrt{N} r^{m+1}_i$ with probability at most

$$2 \exp\left(-\frac{\rho^4 M}{32} \right).$$

A zero row is estimated exactly. The bound is uniform in the retained row, so it remains valid after averaging over the later lists.

Let A be the event that all these row estimates, at all stages and all 2^d inputs, satisfy their deviation bounds. At the last hidden layer there is one output row; at every other stage there are M retained rows. Consequently

$$\Pr(A^c) \le 2^{d+1} LM \exp(-\rho^4 M/32) \le \frac{1}{100}. \tag{9}$$

For completeness, put $a = d + \log L + \log(1/\rho)$ and $M_0 = \lceil 256 \rho^{-4} a \rceil$. Then $M_0 \le 257 \rho^{-4} a$, and $u \exp(-\rho^4 u/32)$ decreases for $u \ge M_0$. Substituting M_0 and using $\log a \le a - 1$ bounds the logarithm of the right side before its final inequality by

$$-(7 - \log 2)d - 6 \log L - 3 \log(1/\rho) + \log 514 - 1 \le -\log 100.$$

This also proves the bound when M was enlarged to be at least $9K$.

Next define the total retained row mass using the *original* row norms:

$$R = N \sum_{s=2}^{L-1} \sum_{a=1}^{M} (r^s_{I_s(a)})^2 + N \|W^L\|_2^2.$$

Its expectation is

$$\mathbb{E}R = M \sum_{s=2}^{L-1} \|W^s\|_F^2 + N \|W^L\|_2^2 \le MK.$$

By Markov's inequality the event $B = \{R \le 8MK\}$ has probability at least $7/8$. In particular, on B the retained row mass at each individual stage is at most $8MK$.

Finally, direct calculation gives

$$\begin{aligned}
\mathbb{E}\left[\frac{d}{M}\|\widetilde{W}^1\|_F^2\right] &= \frac{d}{N}\|W^1\|_F^2, \\
\mathbb{E}\|\widetilde{W}^s\|_F^2 &= \|\overline{W}^s\|_F^2 \qquad\qquad (2 \le s \le L-1), \\
\mathbb{E}\left[M\|\widetilde{W}^L\|_2^2\right] &= N\|\overline{W}^L\|_2^2.
\end{aligned}$$

For an internal matrix, independence of the two adjacent layers' index lists makes each index pair uniform on $\{1,\dots,N\}^2$; the factor $(N/M)^2$ is canceled by the M^2/N^2 sampling factor. Independence among different matrix entries is unnecessary. Thus $\mathbb{E}\kappa_M(\widetilde{W}) \le K$, and the event $C = \{\kappa_M(\widetilde{W}) \le 4K\}$ has probability at least $3/4$. On C every sampled internal matrix has operator norm at most $2\sqrt{K}$, and the sampled readout has norm at most $2\sqrt{K/M}$.

The events need not be independent:

$$\Pr(A \cap B \cap C) \ge 1 - \frac{1}{100} - \frac{1}{8} - \frac{1}{4} > 0.$$

Fix a collection of lists in their intersection.

Reverse-order telescoping. Let F_m be the network with hidden layers $m,\dots,L-1$ sampled and layers $1,\dots,m-1$ still original and truncated. Incoming weights to a sampled layer select its rows, and outgoing weights from a sampled layer are multiplied by N/M. This defines the intermediate networks with different widths in different layers. In particular, F_L is the truncated network and F_1 is the fully sampled network.

When comparing F_{m+1} and F_m, layer m computes the original truncated neuron functions before selection. The only new discrepancy is in the sums into the retained neurons of layer $m+1$. For $m = L-1$, event A bounds the output discrepancy directly by $(\rho/2)\sqrt{K} \le \eta/2$. For $m < L-1$, put $t = L-m$. Events A and B bound the Euclidean discrepancy in layer $m+1$'s preactivation by $(\rho/2)\sqrt{8MK}$. All matrices *after* layer $m+1$ already equal their final sampled matrices, so their operator norms are controlled by C. Propagating the discrepancy therefore changes the output by at most

$$\begin{aligned}
\frac{\rho}{2}\sqrt{8MK}\,(2\sqrt{K})^{t-2}2\sqrt{K/M} &= \rho\, 2^{t-1/2}K^{t/2} \\
&\le \eta\, 2^{-3/2} \le \frac{\eta}{2}, \qquad 2 \le t \le L-1.
\end{aligned}$$

There are $L-1$ sampling stages, contributing at most $1/24$ in total. Adding the truncation error proves the approximation bound in (8); event C proves its norm bound.

If the original network has strict margin $1/2$, the sampled network has strict margin $5/12$. Multiplying its linear readout by $3/2$ restores strict margin $1/2$ and multiplies the weighted norm by at most $9/4$. At width M the resulting entrywise bounds are

$$|W^1_{aj}| \le \sqrt{9KM/d}, \qquad |W^s_{ab}| \le 3\sqrt{K}\ (1 < s < L), \qquad |W^L_b| \le 3\sqrt{K/M}.$$

Each is at most M because $d \ge 1$ and $M \ge 9K \ge 9$. The definition of C_L now applies. Finally, substituting (7) gives $\rho^{-4} = 12^4(L-1)^4 16^L K^{2(L-1)}$, which establishes the stated polynomial dependence at fixed depth. □

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